

Critical mathematical referee report

On “A C^1 Counterexample to the Entropy Conjecture”

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Overall recommendation

Major revision. The chief requirements are a primary-source comparison with Pugh (1975), a corrected account of the distinction between *singular* and *nonsingular* C^1 endomorphisms in light of Hayashi (2020), and a clear treatment of Section 4’s standard entropy ingredients. The precise dimensional statement also requires correction from an implicit $q \geq 1$ to the actual $q \geq 2$ in this construction. These are failures of priority documentation and mathematical framing, not established fatal proof errors. Rejection on grounds of a previously known C^1 theorem or a false central calculation would be unsupported by the sources inspected; conversely, acceptance without the primary historical comparison would leave an avoidable priority question unresolved.

A. Major originality, attribution, and missing-reference concerns

A1. The earliest directly relevant published counterexample discussion is absent from the bibliography

Severity: Major attribution/history; unresolved mathematical-priority comparison.
Confidence: Verified omission; possible overlap of construction mechanisms only.
Manuscript location: Introduction, p. 1, paragraph beginning “Shub’s entropy conjecture”; the bibliography, p. 15..

The manuscript explicitly refers to earlier Lipschitz counterexamples “discussed by Pugh”, but cites Shub’s 2007 ICM survey [10] rather than the original account:

Charles C. Pugh, “On the entropy conjecture: a report on conversations among R. Bowen, M. Hirsch, A. Manning, C. Pugh, B. Sanderson, M. Shub and R. Williams,” in A. Manning (ed.), *Dynamical Systems—Warwick 1974*, Lecture Notes in Mathematics **468**, Springer, 1975, pp. 257–261; [doi:10.1007/BFb0082628](https://doi.org/10.1007/BFb0082628).

Springer’s chapter record verifies the authorship, title, original volume and pagination. Shub’s ICM paper [10, p. 4] explicitly distinguishes two prior Lipschitz settings: endomorphisms already on S^2 , and piecewise-linear/Lipschitz homeomorphisms in

dimension four or higher. The former shares the *logical pattern* of degree/homology growth coexisting with zero entropy; the latter addresses invertibility in a weaker regularity class. Neither, on the inspected evidence, establishes the existence of a C^1 example with the manuscript’s finite-dimensional clock-register mechanism. A Lipschitz example cannot be relabeled C^1 without a differentiability argument, and the current audit has found no verified such smoothing in Pugh.

The issue is that the paper presents a potentially major regularity upgrade while omitting the primary source that needs to be compared most closely. In particular, it gives no mathematical comparison of how its clock slowdown, delayed angular forcing, and auxiliary coordinates overcome the failure of C^1 regularity in the older examples. The original five-page chapter is behind an access-controlled Springer preview; consequently, whether Pugh contains an antecedent of *any specific forcing or timing ingredient* remains **unverified** and should not be implied.

Required action: Cite Pugh (1975) in the introduction and carry out a primary-source, construction-level comparison: manifolds and dimensions, regularity obstruction, dynamical trapping, homological action, and whether a smoothing mechanism was already proposed. State explicitly what mathematical ingredient is additional, without presenting a similarity in overall outcome as identity of constructions.

A2. Omission of the nonsingular-endomorphism theorem obscures the decisive singularity hypothesis

Severity: Substantial contextual omission. **Confidence:** Verified omission and scope of cited external result; verified structural distinction in the submitted formula. **Manuscript location:** Introduction, pp. 1–2 (survey of positive C^1 results and assertion that finite regularity, noninvertibility, and intermediate homology distinguish the example); equation (10), p. 4; reset description, p. 5..

An important positive result is missing:

Shuhei Hayashi, “A forward Ergodic Closing Lemma and the Entropy Conjecture for nonsingular endomorphisms away from tangencies,” *Discrete and Continuous Dynamical Systems* **40** (2020), no. 4, 2285–2313; doi: [10.3934/dcds.2020114](https://doi.org/10.3934/dcds.2020114).

The journal’s published abstract states that the entropy conjecture holds for nonsingular C^1 endomorphisms away from homoclinic tangencies. It extends the Liao-Viana-Yang diffeomorphism result already cited as [3]. This is directly relevant to the manuscript’s emphasis on *noninvertibility*: the distinction required to situate the construction is not merely noninvertibility, but the ability to use genuinely *singular* maps. Specifically, throughout the open reset strip $2 < t < 5$ modulo 100, equation (10) reduces to

$$f(t, z, v_1, \dots, v_q) = (t + 1, z^2, 0, \dots, 0).$$

The derivative annihilates every tangent vector in the q register directions on this open set. Thus f is far from a nonsingular endomorphism. Neither the abstract

nor the introduction emphasizes this strong degeneracy, despite its essential role in erasing the previous passage’s state.

This does *not* mean Hayashi’s theorem applies or contradicts the manuscript: it does not apply to this singular map, and the theorem also assumes avoidance of tangencies. The defect is one of describing the mathematical boundary of the negative result and crediting closely relevant later work in the noninvertible C^1 category.

Required action: Cite Hayashi (2020), state that the constructed map has an open critical set and collapses whole register fibers, and distinguish singularity from non-invertibility. Avoid describing “noninvertibility” alone as the pivotal regularity-class difference.

A3. Lemma 4.1 is a standard consequence of entropy theory rather than an independent new localization principle

Severity: Secondary priority/exposition issue. **Confidence:** Verified mathematical reduction; strongly supported historical attribution to Bowen. **Manuscript location:** Section 4, pp. 11–12, “Lemma 4.1 (Entropy localization)”, especially the two-page exceptional-block argument..

The stated implication

$$F(Z) \subset Z, \quad d(F^n x, Z) \rightarrow 0 \ (\forall x), \quad h_{\text{top}}(F|_Z) = 0 \quad \implies \quad h_{\text{top}}(F) = 0$$

is an immediate application of the variational principle. In fact, if μ is an invariant Borel probability measure and $\varphi(x) = d(x, Z)$, invariance and bounded convergence imply

$$\int \varphi d\mu = \int \varphi(F^n x) d\mu(x) \longrightarrow 0.$$

Thus $\mu(Z) = 1$, and $h_\mu(F) \leq h_{\text{top}}(F|_Z) = 0$; the variational principle then gives the assertion. The manuscript itself explicitly mentions this route on p. 11, citing [6], so it has *not* falsely asserted that the conclusion was previously unknown. Nevertheless, the label “Entropy localization” and the long standalone proof may make an already established formal consequence look like a principal additional ingredient.

A related classical result is Bowen’s theorem that, on a compact metric space, entropy is carried by the nonwandering set: Rufus Bowen, “Topological entropy and Axiom A,” in *Global Analysis*, Proceedings of Symposia in Pure Mathematics **14**, AMS, 1970, pp. 23–41, MR0262459. The exact nonwandering-set equality is confirmed in a survey of topological entropy; Bowen’s original article was located bibliographically but was not read line-by-line. The variational-principle reduction above, which is already indicated in the manuscript, provides the exact matching argument without any need to assert that $\Omega(F) \subset Z$.

Required action: Introduce Lemma 4.1 as a standard consequence of the variational principle, cite Bowen (1970) as appropriate background, and either replace the two-page proof by the invariant-measure argument or clarify why the exceptional-block refinement is needed later. No priority claim should attach to the lemma as such.

A4. Homological/chain-recurrence antecedents warrant a narrower historical explanation

Severity: Secondary bibliographic and framing issue. **Confidence:** Verified bibliographic existence; no exact overlap established. **Manuscript location:** Introduction, p. 1; construction-and-proof paragraph, p. 2; homological calculation, Section 5..

The manuscript does not mention M. Shub and D. Sullivan, “Homology theory and dynamical systems,” *Topology* **14** (1975), 109–132, doi:10.1016/0040-9383(75)90022-1, or D. Fried and M. Shub, “Entropy, linearity and chain-recurrence,” *Publications Mathématiques de l’IHÉS* **50** (1979), 203–214, doi:10.1007/BF02684774. The latter paper discusses filtrations, chain recurrence, entropy, and induced homology actions. This material is germane to the manuscript’s contrast between homological growth and entropy supported on recurrent dynamics; it does *not* imply that the special clock-register example was anticipated, nor do its cited diffeomorphism theorems settle the singular-map construction.

Required action: Add a short historically accurate comparison with the literature on filtrations and chain recurrence, with citations if the argument is presented as a broader conceptual interpretation. Do not claim that either paper proves the present counterexample.

D. Terminology and artificial conceptual packaging

D1. The entropy-argument terminology should disclose its standard reformulation

Severity: Minor to substantial exposition issue. **Confidence:** Verified standard reduction. **Manuscript location:** Section 4, Lemmas 4.1–4.2; Introduction, p. 2, sentence about the direct entropy argument..

There are two separate standard normalizations behind the named devices. First, “entropy localization” means support of invariant probability measures on an attracting invariant subset plus the usual variational principle; see A3. Second, the “register dynamics” of Lemma 4.2, after the first image in a disk, is a factor of a product map

$$[0, A] \times S^1 \longrightarrow [0, A] \times S^1, \quad (r, u) \longmapsto (R(r), u),$$

under the polar-coordinate quotient $(r, u) \mapsto ru$. The relevant established concepts are *monotone interval dynamics*, *topological factors*, and *the product rule for entropy*. A nondecreasing continuous interval map has zero entropy, and the frozen angular factor has zero entropy. The manuscript instead devotes most of pp. 12–13 to a bespoke itinerary count. The count may serve an expository purpose, but no distinction from these standard reductions is identified.

Required action: Replace “Entropy localization” with “Variational-principle localization” or state explicitly that it is standard; introduce Lemma 4.2 as a monotone-interval-factor argument and cite a standard entropy text (e.g. P. Walters, *An Intro-*

duction to Ergodic Theory, Springer GTM 79, 1982, Chapters 7–8, doi:10.1007/978-1-4612-5775-2). If the longer elementary proof is retained, do not assign it novelty or conceptual priority.

Terminological caution. “Clock”, “register”, and “prediction” are defined operationally in this manuscript. This audit found no basis for branding those terms **AI slop** or inferring AI authorship merely from the metaphors. Similarly, the adjective “localization” is not, by itself, an invented mathematical notion. No categorical AI-authorship or plagiarism allegation is supported by the inspected evidence.

E. Other mathematical and expository corrections

1. **The advertised dimension bound conceals a restriction forced by the construction.** **Severity:** Minor but mathematically definite. **Confidence:** Verified. **Manuscript location:** Theorem 1.1, p. 2; the plateau cover (4), p. 3; the sentence “ $\dim M = 2q + 3 \geq 5$ ”, p. 2..

Equation (4) requires $[76, 77]$ to be covered by plateaus on which some $\beta_\ell = 1$, while each β_ℓ is supported in an interval J_ℓ of length *strictly less than one*. If $q = 1$, continuity of the single index choice implies $\beta_1 \equiv 1$ on $[76, 77]$, so its support contains an interval of length one, contradicting the condition on J_1 . Consequently $q \geq 2$, and this actual construction satisfies

$$\dim M = 2q + 3 \geq 7,$$

not merely the uninformative ≥ 5 displayed in the manuscript. The printed inequality is literally true, so it is not a counterexample to Theorem 1.1; nevertheless, in a subject where the smallest possible dimension is part of the mathematical landscape, the presentation obscures the actual restriction.

Required action: Write $q \geq 2$ and $\dim M \geq 7$, or state explicitly that dimension five is not obtained by the proposed plateau cover.

2. **The singularity of the reset mechanism is undercharacterized.** **Severity:** Substantial scope clarification. **Confidence:** Verified. **Manuscript location:** Equation (10), pp. 4–5; discussion of noninvertibility in the proof of Theorem 1.1, p. 15..

The final paragraph proves noninvertibility by observing that register coordinates near ∞ are collapsed. This understates what occurs: the open reset region $2 < t < 5$ collapses *all* register variables, independent of their original values and of z . The map therefore has a large rank-deficient open set and cannot lie in a nonsingular endomorphism class. This is important when comparing the manuscript with modern positive results; see A2.

Required action: State the stronger degeneracy explicitly at the end of Theorem 1.1 and in the abstract or introduction.

3. **The claimed contribution of the entropy section needs sharper separation from standard tools.** **Severity:** Minor. **Confidence:** Verified. **Manuscript location:** Section 4, pp. 11–14; proof-strategy paragraph, p. 2..

The discussion gives the impression that the compactness-based exceptional-block estimate and the radial-itinerary count are separate conceptual advances, even though the formal entropy conclusion reduces to invariant-measure support and monotone radial dynamics. Their inclusion may be justified as self-contained proofs, but the presentation should not let them distract from the novel part whose originality requires checking, namely the C^1 anticipatory forcing series and its delay estimates.

Required action: Reorganize the section to distinguish established entropy reductions from the map-specific construction, and shorten or demote the auxiliary proofs where appropriate.

F. Annotated literature and priority audit

Search date: 8 October 2026. Bibliographic and publisher searches covered work under the terms *Shub entropy conjecture*, C^1 endomorphism, nonsingular endomorphism, Lipschitz counterexample, homology and entropy, nonwandering set entropy, chain recurrence, and corresponding older and newer formulations. Journal records, archived articles, preprints, a lecture handout, bibliographic surveys, and mathematical-forum discussions were screened. Accessible full primary mathematics was checked selectively, rather than all sources exhaustively. The status entries below concern specific mathematical overlaps, not generic topic similarity.

F1. Principal antecedents and substantial restrictions

1. **Pugh (1975), *On the entropy conjecture*.** Full citation and DOI in A1. *Cited?* No. *Access:* Springer primary chapter metadata and citation list; Pugh’s complete argument not inspected. *Status:* **Verified** historical priority for the original discussion, **possible overlap** of construction mechanism. *Relevance:* Most important missing historical reference for the previous Lipschitz examples. *Action:* Mandatory citation and direct comparison; no justified inference of copying or direct C^1 anticipation.
2. **M. Shub (2007), “All, most, some differentiable dynamical systems,” *Proceedings of the ICM 2006*, vol. III, pp. 99-120.** [Author-accessiblePDF](#); see p. 4. *Cited?* Yes, as [10]. *Access:* PDF directly inspected, including its statement separating S^2 Lipschitz endomorphisms from higher-dimensional PL homeomorphisms. *Status:* **Verified** related examples and regularity distinction, not the present C^1 construction. *Action:* Retain, but supplement with Pugh’s primary source.
3. **S. Hayashi (2020), “A forward Ergodic Closing Lemma and the Entropy Conjecture for nonsingular endomorphisms away from tangencies,” *Discrete Contin. Dyn. Syst.* 40(4), 2285-2313.** [DOI](#). *Cited?* No. *Access:* Published journal abstract and bibliographic record inspected; theorem details beyond the abstract not checked. *Status:* **Verified** general scope, no theorem-level

subsumption of this singular example. *Action*: Mandatory citation in the comparison of C^1 endomorphism classes.

F2. Related proof techniques and standard entropy theory

1. **R. Bowen (1970), “Topological entropy and Axiom A,” *Global Analysis, Proc. Sympos. Pure Math.* 14, 23-41, MR0262459.** [Institutional record](#); see also [later entropy survey](#). *Cited?* No. *Access*: Original identified bibliographically, theorem statement checked in later survey, not original proof. *Status*: **Strongly supported** antecedent of standard entropy-restriction principles, not a claim that the exact wording of Lemma 4.1 occurs in Bowen. *Action*: Cite as background, or use the already-cited variational principle directly.
2. **P. Walters (1982), *An Introduction to Ergodic Theory*, Springer GTM 79.** [DOI](#). *Cited?* No. *Access*: Bibliography and standard textbook theorem framework only; no assertion of a particular numbered lemma read in the book. *Status*: **Verified** standard background; **no exact manuscript-priority overlap claimed**. *Relevance*: Variational principle, factors and monotone-interval entropy normalizing Section 4. *Action*: Optional background citation if the section is compressed.
3. **M. Misiurewicz (1976), “A short proof of the variational principle for a $\mathbb{Z}_+^N$ action on a compact space,” *Astérisque* 40, 147-157.** *Cited?* Yes, as [6]. *Access*: Manuscript citation verified; historical theorem used, full primary article not rechecked. *Status*: **Verified** applicability of the standard variational-principle argument given in A3. *Action*: Use this existing reference explicitly for Lemma 4.1 instead of creating an impression of novel entropy machinery.

F3. Historical homological and chain-recurrence context

1. **M. Shub and D. Sullivan (1975), “Homology theory and dynamical systems,” *Topology* 14(2), 109-132.** [DOI](#). *Cited?* No. *Access*: Publisher bibliographic record; a limited searchable PDF excerpt, but not a full theorem-level audit. *Status*: **Possible thematic connection**, not verified construction overlap. *Action*: Historical discussion if chain-level homological constraints are discussed.
2. **D. Fried and M. Shub (1979), “Entropy linearity and chain-recurrence,” *Publ. Math. IHÉS* 50, 203-214.** [DOI](#). *Cited?* No. *Access*: Numdam publisher archive and a searchable excerpt from the filtration proposition around p. 207. *Status*: **Verified** related entropy/homology/filtration framework; **not verified** that it contains the present singular forcing mechanism. *Action*: Contextual citation, not adverse priority conclusion.
3. **M. Shub (1974), “Dynamical systems, filtrations and entropy,” *Bull. Amer. Math. Soc.* 80, 27-41.** [DOI](#). *Cited?* Yes, as [9]. *Access*: Verified bibliographically and via historical review, not examined line-by-line. *Status*: **Verified** origin of the relevant conjectural program; no evidence here of the C^1 map constructed in the submission. *Action*: Retain, and anchor the historical comparison in Pugh and Shub’s subsequent account.

F4. Later results delimiting the manifold and map classes

1. **W. Marzantowicz and F. Przytycki (2008), “Entropy conjecture for continuous maps of nilmanifolds,”** *Israel J. Math.* **165**, 349-379. DOI. Cited? No. Access: Published bibliographic record and abstract checked. Status: **Verified** positive theorem on nilmanifolds only, not a proof for the present sphere-product manifold. Action: Optional landscape citation; not necessary to establish the principal difference.
2. **W. Marzantowicz and F. Przytycki (2008), “Estimates of the topological entropy from below for continuous self-maps on some compact manifolds,”** *Discrete Contin. Dyn. Syst.* **21(2)**, 501-512. DOI; arXiv:0705.1610. Cited? No. Access: Journal abstract and preprint description checked. Status: **Verified** scope of the infra-nilmanifold $K(\pi, 1)$ theorem, no overlap with the sphere-product construction. Action: Optional contextual citation only.
3. **G. Liao, M. Viana and J. Yang (2013), “The entropy conjecture for diffeomorphisms away from tangencies,”** *J. Eur. Math. Soc.* **15**, 2043-2060. DOI. Cited? Yes, as [3]. Access: Published article and its introduction examined. Status: **Verified** restricted diffeomorphism theorem; not applicable to singular self-maps. Action: Relate explicitly to Hayashi’s extension, which is currently omitted.

F5. Additional source classes and unresolved priority searches

A 2026 talk handout on smooth parametrizations by David Burguet, [lectureslides](#), mentions the Shub conjecture and Yomdin’s theorem but yielded no verifiable copy of the manuscript’s delayed-forcing construction. A [2024MathOverflowdiscussion](#) concerns polynomial entropy in relation to Shub’s conjecture but does not announce this C^1 counterexample. These items are *not* evidence of mathematical anticipation and are deliberately not promoted into allegations of priority. Searches of broadly indexed talk, thesis, grant and informal channels cannot rule out earlier privately circulated or poorly indexed results. In particular, no comprehensive database of grant proposals or private seminar correspondence was available. The 1975 Pugh chapter remains the most important uncompleted primary-source comparison.

Referee Instructions

Critical Mathematical Referee Report – Comprehensive Instructions

Act as an exceptionally careful, skeptical, and mathematically sophisticated postdoctoral researcher preparing an
 ↳ extensive referee report on the attached mathematical research paper.

Your objective is to identify every substantive problem that could prevent the manuscript from meeting the standards of a
 ↳ serious mathematical research journal.

In particular, investigate:

1. Missing references, inadequate attribution, and previously known mathematical ideas.
2. Artificial novelty concealed by nonstandard terminology or unnecessarily complicated constructions.
3. Mathematical errors, unjustified claims, and gaps in proofs.
4. Lack of mathematical significance or insufficient contribution to current research.
5. AI slop, misleading terminology, and other substantive exposition problems.

****The report must focus exclusively on deficiencies. Do not praise the paper, summarize correct proofs, or explain why its
 ↳ results are significant.****

Be exceptionally thorough in your investigation, but strictly evidence-based in your conclusions.

1. Literature, originality, and attribution – highest priority

Conduct a comprehensive investigation of existing mathematical research related to every substantial theorem,
 ↳ construction, lemma, and proof technique in the manuscript.

Search as broadly and deeply as possible.

1.1 Sources to investigate

Include, but do not limit the investigation to:

- Published research papers, mathematical journals, books, and monographs.
- arXiv preprints, unpublished manuscripts, and institutional repositories.
- Older mathematical literature, including papers using historical terminology.
- MathOverflow, Mathematics Stack Exchange, mathematical blogs, and other mathematical social networks.
- Publicly available grant proposals, funded research descriptions, project summaries, and progress reports.
- Conference proceedings, workshop programs, seminar announcements, and research meeting schedules.
- Talk titles, abstracts, lecture notes, presentation slides, and recorded mathematical talks.
- PhD theses, dissertations, habilitation theses, and research notes.
- University and research institute websites.
- Publicly accessible correspondence, discussions, or other records of mathematical developments where relevant and
 ↳ reliable.

Investigate both recent and historical sources. Do not restrict searches to papers that use the manuscript's terminology.

Where appropriate, follow citation chains backward to earlier sources and forward to subsequent developments.

1.2 Assume substantial overlap may exist

****Adopt as a working hypothesis that at least one significant mathematical idea, construction, argument, or proof
 ↳ technique in the manuscript may already exist somewhere in the literature.****

Actively investigate this possibility.

Do not assume that an elaborate construction is original merely because it looks unusual or technically sophisticated.

In particular, investigate whether:

1. The main theorem has already been proved, stated, or anticipated.
2. The main result follows readily from an existing theorem.
3. An equivalent or stronger counterexample already exists.
4. The essential construction is a modification of a known construction.
5. A central lemma is a known result presented in different language.

6. A complicated argument is equivalent to a standard argument after a change of notation, variables, coordinates, or
 ↳ parametrization.
7. The manuscript combines known results without a sufficiently new mathematical ingredient.
8. An earlier paper contains the essential idea without explicitly stating the manuscript's final conclusion.
9. An unpublished source, seminar abstract, or research announcement already describes the decisive mechanism.

For every substantial overlap, compare the precise mathematical content, including assumptions, conclusions,
 ↳ constructions, and proof mechanisms.

****A difference in notation, terminology, presentation, or technical implementation does not by itself establish
 ↳ mathematical originality.****

1.3 Missing references and attribution

Compare all relevant sources against the manuscript's reference list.

Pay particular attention to missing citations for:

- The principal mathematical idea.
- The central construction or counterexample.
- The key technical mechanism.
- Previously established intermediate results.
- Standard theorems used implicitly.
- Earlier results that substantially limit the manuscript's novelty.
- Existing interpretations of the same mathematical phenomenon.

Identify whether the manuscript cites the earliest relevant source or only later expositions.

For each important missing reference, provide:

1. Complete bibliographic information.
2. A verifiable link, DOI, arXiv identifier, or other reliable locator.
3. The precise part of the manuscript to which the reference is relevant.
4. The relevant theorem, proposition, construction, page, or section in the earlier source, whenever verifiable.
5. A mathematical explanation of the overlap.
6. The seriousness of the omission.
7. The specific citation or historical clarification the authors should add.

Distinguish between an omitted source that merely supplies background and an omitted source that substantially undermines
 ↳ the claimed originality.

****Failure to cite work containing the central mathematical idea, construction, or argument should be treated as a major
 ↳ issue, generally more serious than a routine, repairable proof gap.****

A potentially fatal mathematical error must nevertheless be reported as a major issue.

Do not accuse the authors of plagiarism or intentional copying merely because an earlier mathematical source has been
 ↳ identified. Separate demonstrable overlap, inadequate attribution, priority questions, and evidence of textual
 ↳ copying.

1.4 OpenAI GitHub exclusion

Assume the manuscript is announced or distributed through OpenAI's GitHub mathematics repository.

Do not include that repository's copy of the manuscript as a related-literature discovery, missing reference, provenance
 ↳ concern, or evidence of copying.

Do not discuss the mere existence of the same paper on OpenAI GitHub.

Other mathematically independent sources remain eligible, including independently authored work, even if those sources are
 ↳ hosted on GitHub.

2. Mandatory terminology normalization before originality assessment

****Before deciding whether a mathematical concept, lemma, construction, or theorem is original, translate its terminology
 ↳ into standard mathematical language wherever possible.****

This is a mandatory stage of the literature investigation, not merely a writing-style check.

AI-generated mathematical manuscripts may use invented terminology, unconventional formulations, or unnecessarily elaborate constructions to describe established mathematical concepts.

Such language can conceal substantial overlap with existing literature.

2.1 Normalization procedure

For every important concept or technique:

1. Extract its precise mathematical definition.
2. Identify its essential mathematical content independently of the manuscript's terminology.
3. Search for the corresponding standard terminology in authoritative mathematical literature.
4. Identify relevant established synonyms and historical terminology.
5. Reformulate the manuscript's claims using that terminology.
6. Determine whether the reformulation is mathematically equivalent to a known result, a special case, a direct corollary, or a modest adaptation.
7. Search existing literature using both the original and normalized formulations.
8. Compare the underlying mathematical mechanisms rather than the wording.
9. Preserve the original terminology as an additional search term, since identical unusual language in an earlier source may itself be relevant.

Where exact equivalence is claimed, verify all assumptions and conclusions. If two concepts overlap only partially, explain the precise relationship.

Do not force a concept into an existing category unless the correspondence is mathematically justified.

2.2 Detecting AI slop

Explicitly identify and report **AI slop** in mathematical terminology and exposition.

Investigate the following forms:

- **Invented terminology:** New names for established mathematical concepts without a substantive mathematical distinction.
- **Pseudo-technical jargon:** Expressions that sound sophisticated but have no precise or recognized mathematical meaning.
- **Unnecessary conceptual packaging:** Artificially named principles or constructions that merely restate elementary observations.
- **Disguised standard results:** Familiar lemmas or techniques presented under unfamiliar names.
- **Terminological misuse:** Established mathematical terms used with incompatible or misleading meanings.
- **Artificial complexity:** Elaborate definitions or technical machinery that obscure a substantially simpler standard formulation.
- **Artificial novelty:** Unconventional terminology or presentation that creates a misleading impression of mathematical originality.

For every confirmed instance, report:

1. The problematic expression and its exact location.
2. The mathematical concept it actually describes.
3. The corresponding standard terminology, if one exists.
4. A reliable mathematical reference supporting that terminology.
5. Why the manuscript's expression is unnecessary, misleading, or inappropriate.
6. A specific recommended replacement.

Use the explicit label **AI slop** for substantiated instances.

Do not classify legitimate new terminology as AI slop merely because it is unfamiliar. Do not infer AI authorship from unusual terminology alone.

2.3 Reassess originality after normalization

After translating the concepts into standard terminology, repeat the relevant literature searches.

If normalization reveals that a supposedly new result or method already exists, report both:

- The terminological problem, where justified.
- The more serious mathematical originality or attribution problem.

The originality assessment must be based on mathematical meaning, not verbal novelty.

Terminology normalization must occur before the final assessment of research significance.

3. Mathematical correctness – report errors only

Examine the entire manuscript rigorously, including all theorems, propositions, lemmas, constructions, examples, $\hookrightarrow$ definitions, and proofs.

Do not merely check whether the arguments appear plausible. Independently test the most delicate steps.

Investigate:

- False mathematical statements.
- Unjustified assumptions.
- Missing hypotheses.
- Circular reasoning.
- Incorrect inequalities or estimates.
- Invalid limiting arguments.
- Uniformity and compactness issues.
- Incorrect interchange of limits, derivatives, integrals, or infinite sums.
- Unjustified convergence or regularity claims.
- Errors involving exceptional cases or boundary behavior.
- Invalid applications of established theorems.
- Quantifier errors.
- Hidden dependence of constants on supposedly independent parameters.
- Incomplete constructions.
- Problems involving topology, differentiability, measure theory, geometry, or other relevant subjects.

Where appropriate, test assertions using alternative mathematical arguments, explicit examples, symbolic calculations, or $\hookrightarrow$ numerical experiments.

For every genuine or suspected problem:

1. Identify the exact location.
2. State the problematic claim.
3. Explain precisely why the stated argument is invalid, incomplete, or insufficiently justified.
4. Supply a counterexample or mathematical obstruction whenever possible.
5. Explain whether the issue affects only a local statement or undermines the main theorem.
6. Distinguish a missing explanation from a genuine mathematical gap.
7. Suggest the minimum correction or additional hypothesis needed, when identifiable.

Classify the issue as minor, substantial, or potentially fatal.

****Do not include correct mathematics merely to demonstrate that it has been checked.****

Do not summarize valid proofs, praise technically successful arguments, or include a section listing correct results.

Only reproduce mathematical calculations needed to demonstrate a specific error or deficiency.

If no mathematical error is found in a section, omit discussion of that section.

4. Mathematical significance – report deficiencies only

Assess whether the manuscript makes a meaningful contribution relative to current mathematical research.

A mathematically correct result need not be significant.

In particular, investigate whether:

1. The main theorem is already known or follows easily from existing results.
2. The result addresses a weak, artificial, or peripheral formulation of an important mathematical problem.
3. The counterexample merely exploits the absence of a natural assumption.
4. The construction reveals a genuinely informative mathematical mechanism.
5. The construction is unnecessarily complicated relative to its conclusion.

6. The result improves our understanding of the underlying mathematical equation, inequality, conjecture, or dynamical phenomenon.
7. The main theorem has consequences beyond the specific artificial example.
8. The work meaningfully advances current research directions.
9. The claimed contribution is disproportionately small relative to the technical machinery used.
10. The abstract or introduction overstates what the result establishes.

Special scrutiny of counterexamples

****A counterexample may disprove a formally stated conjecture without significantly improving mathematical understanding of the underlying problem.****

Do not confuse logical refutation with conceptual progress.

Investigate whether the counterexample identifies an important obstruction or merely exploits a technical loophole.

Determine whether the example sheds light on the original motivation for the conjecture, its natural variants, or the reasons the inequality or theorem might be expected to hold.

A counterexample that settles only an unusually broad or technically weak formulation should not automatically be described as a major advance.

If the result has insufficient conceptual significance, explain this as a major issue with precise reference to the relevant research landscape.

****If the paper makes a meaningful contribution and there is no significance problem, do not include a positive assessment. Omit the significance section entirely.****

5. Exposition, presentation, and bibliographic accuracy

Identify substantive writing deficiencies, including:

- Incorrect or incomplete historical claims.
- Misleading statements of novelty.
- Nonstandard terminology where standard terminology exists.
- AI slop.
- Unnecessarily complicated descriptions of standard concepts.
- Ambiguous mathematical statements.
- Missing definitions or hypotheses.
- Inconsistent notation.
- Misleading theorem names.
- Incorrect citations.
- Inaccurate bibliographic information.
- Arguments whose presentation conceals a genuine logical dependency.
- Unnecessary technical machinery that obscures the main mathematics.
- Grammatical or structural problems that materially affect mathematical clarity.

For each issue, give its location and a concrete recommended correction.

Do not include minor stylistic preferences unless they materially improve clarity, accuracy, or mathematical communication.

Avoid general praise and positive exposition commentary.

6. Verification and evidence standards

The report must be exceptionally critical but must never manufacture evidence.

For every external reference:

- Verify that the source exists.
- Verify its authorship and bibliographic information.
- Verify its mathematical relevance.
- Check the exact statement or construction whenever the source is accessible.
- Distinguish directly inspected sources from sources known only through abstracts, citations, or secondary reports.
- Supply reliable links and precise locations where available.

- Distinguish mathematical equivalence from merely suggestive similarity.
- Do not invent theorem numbers, page references, quotations, publication dates, or priority claims.

Record the search date and clearly identify any important limitations of the literature investigation.

A source found through a search snippet alone must not be treated as proof of an exact mathematical overlap.

If a grant proposal or talk abstract suggests that an idea was known earlier, explain what the available document actually establishes and what remains unverified.

Do not infer that a full theorem was proved merely because an earlier abstract announced a related research direction.

Where uncertainty remains, use explicit categories:

- **Verified:** Direct documentary or mathematical evidence establishes the claim.
- **Strongly supported:** Substantial evidence exists, but some details remain unverified.
- **Possible overlap:** The relationship is plausible but not established.

Do not present speculative similarities as confirmed originality failures.

Nevertheless, actively investigate plausible leads rather than dismissing them prematurely.

The goal is maximum detection of genuine problems with minimum false accusations.

7. Ranking the problems

Prioritize substantive findings according to their effect on the manuscript.

Highest priority

- An uncited earlier source containing the central result, construction, or decisive argument.
- A previously known theorem that substantially subsumes the manuscript's main result.
- A fatal mathematical error invalidating the central theorem.
- A fundamental lack of originality concealed by terminology, presentation, or technical complications.

Major concerns

- Serious failure to acknowledge essential prior research.
- Substantial proof gaps affecting principal conclusions.
- A lack of meaningful mathematical contribution.
- A misleading description of the conjecture or problem purportedly resolved.
- A key technical argument that reproduces an existing argument without appropriate acknowledgment.

Secondary concerns

- Repairable local mathematical gaps.
- Nonstandard terminology and substantiated AI slop.
- Missing contextual references that do not affect priority.
- Inaccurate notation, exposition, or bibliographic details.

This ordering is intended to distinguish the seriousness of findings, not to assert that all bibliographic issues outweigh all mathematical errors.

For each finding, provide:

- Severity.
- Confidence level.
- Precise location.
- Evidence.
- Required action.

Focus the discussion on the most consequential issues rather than giving equal weight to every observation.

8. Required referee report structure

Prepare the report in the following format.

A. Major originality, attribution, and missing-reference concerns

Present the most serious issues first.

For each major overlap, explain the precise mathematical correspondence with earlier work.

Highlight references absent from the manuscript.

Distinguish known results, substantial adaptations, and possible overlap requiring further verification.

B. Mathematical errors and proof gaps

Present identified mathematical defects in decreasing order of seriousness.

Include the mathematical details necessary to substantiate each criticism.

Do not discuss correct arguments.

C. Insufficient mathematical significance

Include this section only when there is a substantive concern.

Explain exactly why the paper's main result may not constitute meaningful progress.

For counterexamples, distinguish resolving a formal statement from improving the mathematical understanding of the
 $\hookrightarrow$ motivating problem.

Do not include a positive significance assessment.

D. AI slop and terminology problems

Identify nonstandard, misleading, or artificially complicated terminology.

Give standard replacements and supporting references.

Explain when normalization reveals a previously known mathematical concept or result.

Do not repeat the full originality discussion; cross-reference Section A where appropriate.

E. Other mathematical and expository corrections

Provide a numbered list of additional issues that require correction.

Avoid trivial stylistic complaints.

F. Annotated literature and priority audit

Provide a detailed list of relevant sources discovered during the investigation.

Organize the sources into:

1. Substantial prior results or constructions.
2. Related proof techniques and standard mathematical machinery.
3. Historical antecedents.
4. Recent research and developments.
5. Relevant unpublished sources, mathematical social networks, grants, seminars, talks, abstracts, and theses.

For each source, record:

- Complete bibliographic information.
- Link or identifier.
- Exact mathematical relevance.
- Whether the source is already cited in the manuscript.
- Whether the source was directly inspected.
- Whether the overlap is verified, strongly supported, or merely possible.
- Whether an additional citation or clarification is required.

Do not inflate the bibliography with marginally related references. Prioritize sources that could affect originality,
 ↪ attribution, correctness, or significance.

G. Overall recommendation

Give a definite recommendation:

- Accept.
- Minor revision.
- Major revision.
- Reject.

Justify the recommendation entirely in terms of identified deficiencies.

Explain which issues are essential to resolve before publication.

Do not balance criticism with unnecessary praise or summarize the successful aspects of the paper.

If no substantive problems are found in a category, omit that category rather than filling it with positive commentary.

9. Research workflow

Follow this order as closely as possible:

Stage 1 – Mathematical extraction

Identify the main claims, essential constructions, decisive lemmas, and key proof mechanisms.

Use this information internally; do not include a routine summary in the final report.

Stage 2 – Terminology normalization

Translate nonstandard expressions into established mathematical terminology where justified.

Identify possible AI slop.

Stage 3 – Broad literature discovery

Search using original terminology, standard terminology, synonyms, historical terminology, equivalent formulations, and
 ↪ characteristic mathematical mechanisms.

Search all relevant source categories listed above.

Stage 4 – Mathematical comparison

Read the most relevant sources closely.

Compare assumptions, conclusions, constructions, and arguments. Investigate whether apparent differences are merely
 ↪ notational or presentational.

Stage 5 – Priority and attribution audit

Compare the discovered literature with the bibliography and historical claims of the manuscript.

Identify consequential missing citations and originality problems.

Stage 6 – Independent mathematical verification

Check the main arguments rigorously and look for mathematical errors.

Pay particular attention to technically delicate steps and arguments whose validity is essential to the claimed novelty.

Stage 7 – Significance assessment

Evaluate the mathematical contribution in light of the normalized concepts and verified earlier literature.

Report only genuine deficiencies.

****Stage 8 – Final critical report****

Present only substantive problems, with evidence, severity classifications, and specific requested corrections.

Do not stop the literature investigation after finding the first relevant reference. Continue searching for earlier or
 ↪ stronger sources that may change the priority assessment.

10. Output requirements

1. Write a comprehensive, technically detailed referee report suitable for a research journal in mathematics.
2. Use precise mathematical language and appropriate LaTeX notation.
3. Identify exact page numbers, theorem numbers, equations, and proof locations whenever possible.
4. Provide verifiable bibliographic references and hyperlinks.
5. Clearly distinguish verified findings from unconfirmed suspicions.
6. Do not include generic praise, positive assessments, or descriptions of correct mathematics.
7. Give particular prominence to substantial missing references and disguised previously known arguments.
8. Explicitly identify confirmed instances of AI slop and supply standard terminology.
9. Treat insufficient mathematical significance as a major concern when supported by the research comparison.
10. Include this entire instruction prompt verbatim, exactly once, in an appendix titled *****Referee Instructions*****,
 ↪ without recursive duplication.
11. Set the author of the referee report to ****ChatGPT-6****.
12. Export the report as a complete, standalone, compilable ****LaTeX (.tex) file****.
13. Also produce a compiled PDF whenever technically possible.
14. Ensure that the bibliography, mathematics, hyperlinks, and document structure compile correctly.
15. Complete as much of the investigation as the available tools and sources permit. Explicitly disclose limitations
 ↪ instead of claiming exhaustive verification without evidence.

Governing principle

****The referee's principal task is to discover what is wrong, previously known, inadequately attributed, unnecessarily
 ↪ disguised, or mathematically insignificant—not to certify what is correct.****

In particular:

- Normalize terminology before assessing originality.
- Look for mathematical substance beneath unfamiliar jargon.
- Investigate whether the decisive ideas already exist in the literature.
- Treat serious missing attribution as a major research-integrity and scholarly-quality concern.
- Report every substantiated mathematical error.
- Criticize lack of significance when warranted, but do not praise significance when it is present.
- Label substantiated artificial mathematical jargon as AI slop.
- Make strong criticisms only when supported by precise mathematical or documentary evidence.

The final report should be rigorous, skeptical, specific, and useful to an expert editor deciding whether the manuscript
 ↪ satisfies high standards of mathematical originality, correctness, attribution, and significance.